

Joey Chung

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EDUCATION

University of Michigan, Ann Arbor

B.S. in Chemistry | GPA: 3.99 / 4.00

Aug. 2025 – Present

University of Wisconsin, Madison

Coursework toward B.S. in Food Science | GPA: 3.96 / 4.00

Sept. 2022 – May 2024

Honors: Dean's List (4 semesters); Norman F. Olson Family Scholarship (2024); Bill Reese Dairy, Deli & Bakery Study Grant (2023); Wisconsin Alumni Association Korean Chapter Scholarship (2022)

PROFESSIONAL EXPERIENCE

DAEYANG Co. Ltd | Research Assistant

Daegu, South Korea | Sept. 2024 – May 2025

Corporate research institute developing thermoelectric materials and modules for energy applications (re-dy.com/en)

- **Surface Engineering & ENi Plating:** Performed medium phosphorus (6 to 8% P) electroless nickel plating on bismuth telluride pellets, including alcohol-based cleaning, pH monitoring and adjustment; proposed and designed a custom titanium holding jig, including technical drawings, to improve plating stability (implementation deferred due to production priorities).
- **Dimensional Inspection & Defect Classification:** Inspected diced thermoelectric legs for dimensional accuracy and surface defects, measuring critical dimensions against design tolerances and classifying defect types in collaboration with the Korea Electrotechnology Research Institute (KERI).
- **SMT Assembly & Wiring Support:** Assisted senior engineers with SMT based assembly of 8×8 thermoelectric modules (128 legs each) by preparing solder paste, setting up mounting frames, loading substrates, and soldering lead wires to module electrodes after reflowing; supported weekly production of 20 to 30 modules, scaling to 50+ during government R&D cycles.
- **QA Screening & Traceability:** Conducted final electrical inspection on finished modules by measuring resistance and voltage, verifying spec ranges, documenting results in Excel for traceability, and flagging nonconforming units for retesting.
- **TEG Performance Testing:** Collected thermoelectric generator performance data across cooling configurations, including forced and natural air cooling, forced and natural water cooling, and antifreeze-based cooling, to support comparative analysis.

Military Service | Auxiliary Police Officer (Sergeant)

Pohang, South Korea | Nov. 2019 – May 2021

Mandatory service under the Ministry of National Defense and the Korean National Police Agency

- **Operational Leadership & Search and Rescue:** Led a squad of 12 officers for 7 months during high pressure field deployments, including protest response and missing person searches across mountainous and coastal terrain; located 3 missing individuals and escorted them to safety, earning an official commendation.
- **Crisis Management & Public Safety:** Managed police lines during civil disturbances, conducted joint counterterrorism drills including chemical threat response procedures, provided driving support to transport personnel and equipment, and maintained station access control while triaging civilian reports.

RESEARCH & TEACHING

Maldonado Group | Undergraduate Research Assistant

UM-Ann Arbor | April 2026 – Present

Advisor: Prof. Stephen Maldonado

- **Research Instrumentation & Surface Analysis:** Conducting materials and surface characterization research using Atomic Layer Deposition (ALD), ellipsometry, and Atomic Force Microscopy (AFM), supporting sample handling, instrument setup, measurement collection, and documentation.

Organic Chemistry Peer Mentor | Academic Support

UW-Madison | Fall 2023 – Spring 2024

- **Instructional Leadership:** Resolved 100+ academic inquiries on Piazza and led interactive sessions simplifying advanced organic chemistry concepts and multistep reaction mechanisms.

LEADERSHIP & GLOBAL ENGAGEMENT

Michigan Rifle Team | Member

UM-Ann Arbor | Sept. 2025 – Present

- **Precision & Team Contribution:** Competed in 3 position and air rifle events (Camp Perry Open, WIRC Championship); created social media photo content and organized team building events.

Food Science Club | Member

UW-Madison | Sept. 2022 – Dec. 2023

- **Industry & Corporate Relations:** Coordinated HR led recruitment sessions with industry representatives and organized collaborative cooking activities to boost member engagement.

International Food and Cultural Engagement | Cultural Outreach

8+ Countries | Jan. 2021 – Present

- **Cultural Diplomacy:** Promoted Korean fusion cuisine (e.g., bulgogi tacos, gochujang infusions) through cooking sessions and festival style events across countries including Spain, Poland, and Mexico.

SKILLS

- **Process & Quality Engineering:** SMT workflow, parameter control, solder paste and surface prep, fixture design, dimensional inspection, defect classification, comparative evaluation, data logging and traceability, Excel reporting, power calculations.
- **Materials & Surface Characterization:** Atomic Layer Deposition (ALD), ellipsometry, Atomic Force Microscopy (AFM).
- **Leadership & Safety:** Safety first execution, checklist readiness, high pressure coordination, clear technical communication.